# SOFT255SL: Software Engineering for the Internet Using Java

This Course work is worth 10% of the module marks.

Quiz Date: 10th December 2016 -TBC

Mode of Assignment: Individual – Practical Test

Module Leader: Manoja Weerasekara

Module Leader Contact Details: manoja@nsbm.lk

Guidelines: Answer all questions Total mark allocation: 100

Time allocation: 1hr 30min

**PART 01: Multiple Choice Questions (Total: 20*2.5= 50 Marks)**

1. In an object model, there are major and minor elements. Which one of the following is true?

a. Abstraction, Encapsulation and Persistence are the major elements.

b. Hierarchy, Concurrency and Typing are the major elements.

c. Abstraction, Encapsulation and Hierarchy are the major elements.

d. Typing is the major element.

2. Which one of the following terms must relate to "polymorphism?"

a. Static allocation.

b. Static typing.

c. Dynamic binding.

d. Dynamic allocation.

3. What is an abstract class?

a. An abstract class is one without any child classes.

b. An abstract class is any parent class with more than one child class.

c. An abstract class is class which cannot be instantiated.

d. abstract class is another name for "base class."

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4. What attributes do all real world objects have?

a. Objects have identity, state, and behavior.

b. Objects have state and behavior.

c. Objects have size and weight.

d. Objects have existence.

5. Examine the following section of code:

String strA;

String strB = new String("Cheese");

How many objects have been created?

a. zero

b. one

c. two

d. three

6. What is said of an operator like "+" when it has several meanings depending on context?

a. overloaded

b. overlorded

c. overworked

d. overlooked

7. What is the effect of giving a class member private access?

a. When a member of a class is declared private it can be used in only one place in a program.

b. When a member of a class is declared private it can be used only in methods that are members of that class.

c. When a member of a class is declared private it can only be used by other private

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members of other classes.

d. When a member of a class is declared private there will be only one instance of it, no matter how many objects are instantiated.

8. Fill in the blank in the following code fragment so that each element of the array is assigned twice the value of its index.

```
int[] array = new int[10];
// scan the array
for ( int index=0; index < array.length; index++ )
{
_____________________
}
```

a. index = 2*index;

b. array[ 2*index ] = 2*index;

c. array[ index ] = 2*array[ index ];

d. array[ index ] = 2*index;

9. Does a subclass inherit both member variables and methods?

a. No—only member variables are inherited.

b. No—only methods are inherited.

c. Yes—both are inherited.

d. Yes—but only one or the other are inherited.

10. Can an abstract class define both abstract methods and non-abstract methods?

a. No—it must have all one or the other.

b. No—it must have all abstract methods.

c. Yes—but the child classes do not inherit the abstract methods.

d. Yes—the child classes inherit both.

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11. Examine the following code:

Vector list = new Vector(10);

list.addElement( new Point(5,12) );

list.addElement( new Point(15,23) );

list.addElement( new Point(62,72) );

After the code has executed, what is the capacity of the Vector? What is its size?

a. 3, 3

b. 3, 10

c. 10, 3

d. 10, 10

12. What happens in a method if an exception is thrown in a try{} block and there is NO MATCHING catch{} block?

a. This is not legal, so the program will not compile.

b. The method throws the exception to its caller, exactly if there were no try{} block.

c. The program halts immediately.

d. The program ignores the exception.

13. Which statement is FALSE about catch {} blocks?

a. There can be several catch{} blocks in a try/catch structure.

b. The catch{} block for a child exception class must PRECEED that of a parent exception class.

c. The catch{} block for a child exception class must FOLLOW that of a parent exception class.

d. If there is no catch{} block there must be a finally{} block.

14. Which statement is FALSE about the try{} block?

a. Some of the statements in a try{} block will never throw an exception.

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b. The statements in a try{} block may throw several types of exception.

c. The try{} block can not contain loops or branches.

d. The try{} block must appear before the catch{} blocks.

15. What will be the result of the expression “a % b” when a and b are of type int and their values are a = -17 and b = -6?

a. 5
b. -5
c. -23
d. 23

16. What would be the output if we execute the expression (1 | 4) + (4 & 2)

a. 5
b. 1
c. 2
d. 8

17. Consider the following code fragment:

System.out.println( 1 + 2 + 3 + 4 + 5 + "-nsbm-" + (6 + 7 + 8 + 9) );

Which one of the following will be displayed when the above fragment is executed?

a. 12345- nsbm -6789
b. 15- nsbm -30
c. 12345- nsbm -30
d. 15- nsbm -6789

18. Consider the following fragment of code

What is the value of k after the following code fragment?

```
int k = 0;

int n = 12

while (k < n)

{

 k = k + 1;}
```

What is the value of k after the following code fragment?

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a. 0
b. 11
c. 12
d. Unknown

19. Consider the following fragment of code

```
int i=10;

int n= i++%5;
```

What are the values of I and n after code is executed?

a. n= 0 and i=10
b. n= 1 and i=10
c. n= 1 and i=11
d. n= 0 and i= 11

20. Which one of the following correctly explains the meaning of overriding?

a. When a method is declared with the same name as another method but with a different set of parameters.
b. When a method is declared in a subclass with the same name, parameters and return type (or a subtype) as in a superclass.
c. When a variable is declared with the same name as another variable in the same class.
d. When a subclass is declared with the same name as its superclass.
e. When a subclass has a variable declared with the same name as a method in a superclass.

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**Part B: Structured Essay Questions (Java). Write down answers on the same paper.**

01. Consider the following interface.

```
public interface SomethingIsWrong {
        void aMethod(int  aValue){
                System.out.println("Hi Friend");
        }
}
```

a. What is incorrect in the above interface? (2 mark)

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b. Fix the interface in the question above. (3 mark)

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c. Is the following interface valid? Why? (4 mark)

```
public interface Marker{
}
```

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d. List main **similarities** between an "Interface" and a "Class" in java. (4 mark)

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e. What is the advantage of having "Interface" concept in java? (2 mark)

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SAMPLE

02. Consider the following class:

```
public class identifyMyParts {
        public static int  x = 7;
        public int  y = 3;
}
```

a. What are **class** variables in java? (3 mark)

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b. Identify the **class variables** in this code snippet. (2 mark)

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c. Identify the **instance variables** in this code snippet. (2 mark)

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d. Write down the statement to **declare a constant and initialize a value**. (2 mark)

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e. What is the output from the following code? (6 mark)

```
double [] data = { 8.0, 10.0, 14.0, 10.0, 8.0 };
double [] smoothed = new double[data. length];
int d = data.length-1;
smoothed [d] = data[d];

for (int i = 0; i < d; i++){
        smoothed [i] = 0.5*(data[i]+data[i+1]);
        System.out.print (smoothed[i] +" , ");
}

System.out.println (smoothed[d]);
```

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SAMPLE

**Part C: Essay Questions (Java). Write down answers on the same paper.**

A thread is a single sequence of execution within a program refers to multiple threads of control within a single program each program can run multiple threads of control within it.

Write a program which will create a named Thread called ("Runner"). Set priority to 6 and run the thread to print thread name ("Runner") 10 times on the console. **(10 Mark)**

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